

2026-09-05 - Handout – Priority Queue (Heap)

Q1 Kth Smallest Element in a Sorted Matrix

Link <https://leetcode.com/problems/kth-smallest-element-in-a-sorted-matrix/description/>

Given an $n \times n$ matrix where each of the rows and columns is sorted in ascending order, return the k^{th} smallest element in the matrix.

Note that it is the k^{th} smallest element **in the sorted order**, not the k^{th} **distinct** element.

You must find a solution with a memory complexity better than $O(n^2)$.

Example 1:

Input: matrix = `[[1,5,9],[10,11,13],[12,13,15]]`, k = 8

Output: 13

Explanation: The elements in the matrix are `[1,5,9,10,11,12,13,13,15]`, and the 8th smallest number is 13

Q2. Task Scheduler

Link <https://leetcode.com/problems/task-scheduler/description/>

You are given an array of CPU tasks, each labeled with a letter from A to Z, and a number n. Each CPU interval can be idle or allow the completion of one task. Tasks can be completed in any order, but there's a constraint: there has to be a gap of **at least** n intervals between two tasks with the same label.

Return the **minimum** number of CPU intervals required to complete all tasks.

Example 1:

Input: tasks = `["A","A","A","B","B","B"]`, n = 2

Output: 8

Explanation: A possible sequence is: A -> B -> idle -> A -> B -> idle -> A -> B.

After completing task A, you must wait two intervals before doing A again. The same applies to task B. In the 3rd interval, neither A nor B can be done, so you idle. By the 4th interval, you can do A again as 2 intervals have passed.

Q3. Minimum Cost To Reach City With Discounts

Link: <https://leetcode.com/problems/minimum-cost-to-reach-city-with-discounts/description/>

A series of highways connect n cities numbered from 0 to $n - 1$. You are given a 2D integer array `highways` where `highways[i] = [city1i, city2i, tolli]` indicates that there is a highway that connects `city1i` and `city2i`, allowing a car to go from `city1i` to `city2i` and vice versa for a cost of `toll`.

You are also given an integer `discounts` which represents the number of discounts you have. You can use a discount to travel across the i th highway for a cost of `tolli / 2` (integer division). Each discount may only be used once, and you can only use at most one discount per highway.

Return the minimum total cost to go from city 0 to city $n - 1$, or -1 if it is not possible to go from city 0 to city $n - 1$.

Input: $n = 5$, `highways = [[0,1,4],[2,1,3],[1,4,11],[3,2,3],[3,4,2]]`, `discounts = 1`

Output: 9

Explanation:

Go from 0 to 1 for a cost of 4.

Go from 1 to 4 and use a discount for a cost of $11 / 2 = 5$.

The minimum cost to go from 0 to 4 is $4 + 5 = 9$.

